In-context learning alters representation of task-related tokens

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★ *Introduction*

Core idea: for classification tasks performed by in-context learning, ***model performs by assigning inputs belongs to different labels onto distinct, different manifolds to form clusters***.

Consequently, at test time model ***reads*** the label by the representation of the question token.

- ***Task settings***
  The classification problem is simple. Inputs are integers, each has label

$$\text{label}(x) = \text{alphabet}(x \bmod 10)$$

The context of (x, label) pairs are given as such:

`1: A, 11: A, ..., 101: A, 2: B, 12: B, ..., 102: B, 3: C, 13: C, ..., 103: C`

The out-of-distribution test case "111" is appended to the context to form the input.

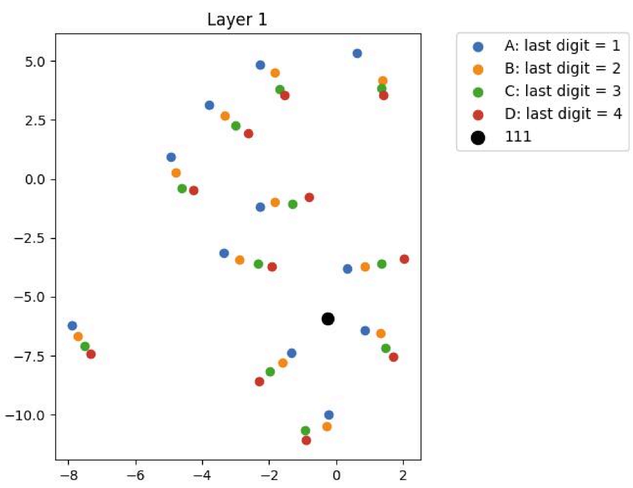
Why choose such input?

A: Because this can verify **if in-context learning DO alters the representation** of input. Input as integers itself has a "STRONG prior" , gain from pre-train. (integers that are close to each other are also close in representation space)

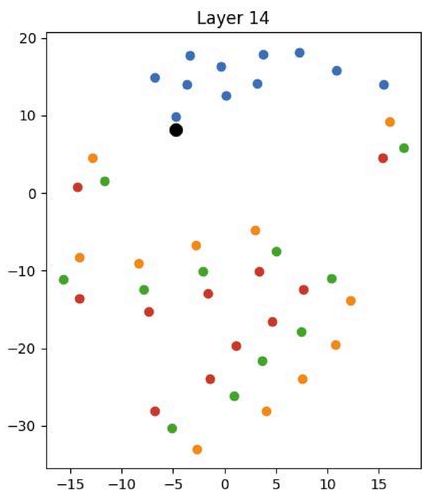
Can model solve this task?
A: Yes. meta-3.1-llama-8b can solve this problem with 24% top1 accuracy.

Dynamics of representation

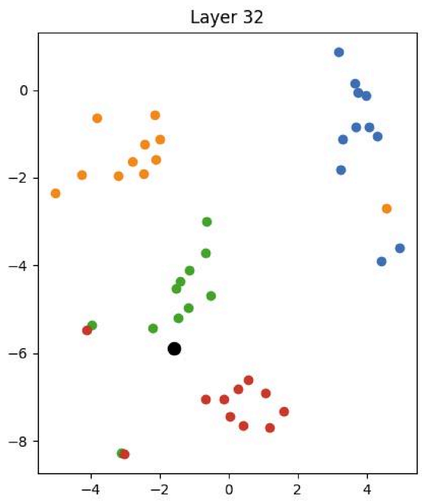
The following plots are t-SNE of the activation of integers tokens: 1, 11, ...



First layer: the map is not learned, representation
learned in pre-train dominates (close number stay together)



Middle layer: compete with pretrain representation, and form a cluster for each label.
In this plot, the question token 111 lies on the correct cluster.



Last layer: well clustered. However
1. Question token does not belong to the correct cluster A.
2. Label A is the top1 prediction given by model, but the plot does not reflect this fact.

★ *Theory*

We propose 3 factors that affect the accuracy of ICL classification

1. How much are manifolds separated?

With the inductive bias that the classification should be able to solved by linear decision boundary, we use SVM with linear kernels to see the upper bound of linear-separability.

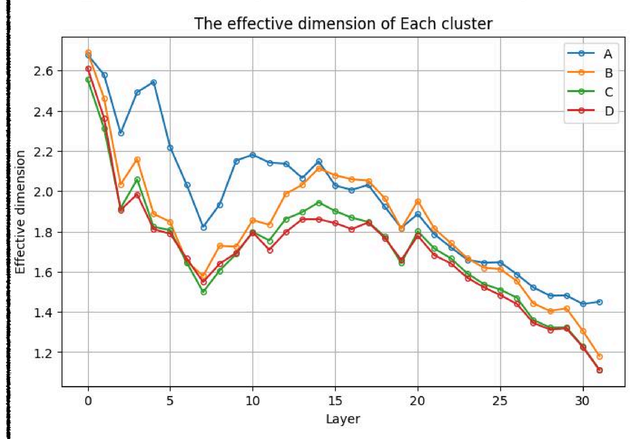
The ***separability of manifolds*** is defined as:

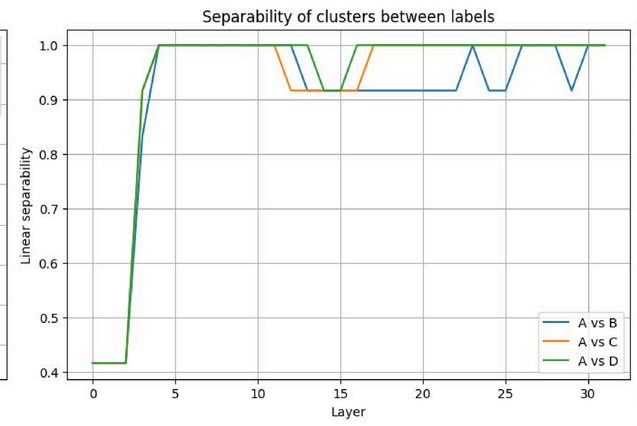
$$S^\star(A) := \sup_{\|u\|=1} \frac{1}{n}\left[\sum_{i\in A}\mathbb{I}(u^\top h_i > 0) + \sum_{i\notin A}\mathbb{I}(u^\top h_i < 0)\right]$$

Instead of train SVM on representation inside the model, we first find the effective dimension of each cluster by

$$\alpha = \frac{\left(\sum_i \sigma_i^2\right)^2}{\sum_i \sigma_i^4}, \quad \text{where } X = U\Sigma V^\top$$

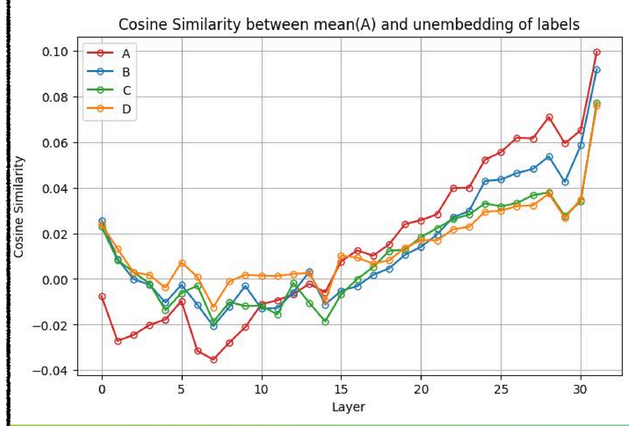
Because in high-dimensional vector space, vectors have high probability to be mutually orthogonal.

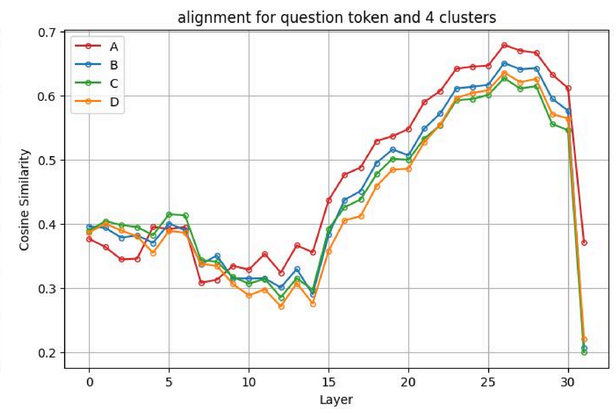




2. How much are the manifolds aligned with the unembedding of label?

Model ***reads the label of each manifold through unembdding.*** Therefore, how much are the manifolds aligned with the unembedding of labels is crucial.



3. Is the question token assigned to the correct manifold?

Model has one more thing to do: assign the question token on the correct label.